

AI-DRIVEN PREDICTIVE & ROBUST QUALITY ENGINEERING

A Quality Engineering Framework for Personalised Content Systems

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01 — CASE OBJECTIVE

Modern content platforms operate at an enormous scale, where user interests, behaviour and expectations continuously change. A quality approach for such an environment cannot depend only on fixed user segments or periodic manual review.

This case study proposes an **AI-driven Quality Engineering** framework that continuously interprets large-scale user signals, identifies emerging quality risks and supports preventive improvement before problems become widespread.

The objective is not simply to recommend what should be improved. The proposed system also considers how an improvement should be implemented, what could go wrong during implementation, and how those risks can be reduced beforehand.

02 — QUALITY PROBLEM

A personalised content system may gradually become less relevant when recommendations fail to adapt to changing interests or repeatedly surface similar content. Three key quality concerns are addressed within this framework:

- **Relevance:** Content may no longer match the user's current interests.
- **Repetition:** Similar content may appear repeatedly, reducing diversity.
- **Adaptability:** The system may respond slowly when behaviour changes.

At Meta's scale, manually creating and updating large user segments becomes inefficient. The proposed approach uses **AI-assisted analysis** of behavioural and feedback signals to identify patterns dynamically.

03 — AI-DRIVEN SIGNAL ANALYSIS

Instead of relying on fixed structures, the system continuously analyses multiple dimensions of user interaction:

- **Explicit Signals:** Likes, follows, feedback and direct user interactions.
- **Behavioural Signals:** Content viewed, skipped, re-watched and engagement duration.
- **Emerging Signals:** Changes in interests and newly developing preference patterns.

An **AI/LLM-based analysis layer** combines these signals to identify meaningful changes at scale without replacing human engineering judgement.

04 — FROM DETECTION TO PREDICTIVE QUALITY

The proposed framework extends beyond reactive problem detection by leveraging **Predictive Quality** methodologies. AI continuously analyses quality signals to identify:

- **Risk Identification:** Potential relevance degradation and repetition patterns.
- **Change Tracking:** Emerging shifts in user behaviour and preferences.
- **Impact Evaluation:** Possible future failures and proposed improvement risks.

This connects naturally with Philip Crosby's **Zero Defects** philosophy: emphasizing prevention rather than correction after failure occurs.

ANALYTICAL SIGNAL PROGRESSION ARCHITECTURE



05 — FMEA-BASED RISK PREDICTION

Before implementing any proposed improvement, potential implementation failures are rigorously evaluated using **Failure Mode and Effects Analysis (FMEA)**. Each failure mode is examined through Severity, Occurrence, and Detection to establish an **RPN-based risk perspective**.

The AI-assisted system supports this analysis by identifying possible consequences, estimating impact areas, and recommending preventive actions for human review. The core operational question shifts from "What should we improve?" to "What could fail while implementing the improvement, and how can we prevent it?"

FRAMEWORK IMPLEMENTATION & ROBUST VALIDATION

06 — VIRTUAL SIMULATION & IMPLEMENTATION

Before exposing a proposed improvement to real users, the system performs a virtual implementation assessment. The AI analyses the change and identifies potential failure modes, cost impact, time impact, productivity impact, and overall implementation risks.

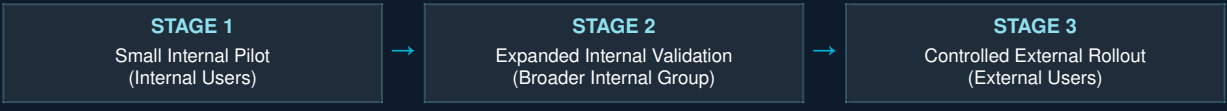
Based on this assessment, an implementation plan is recommended for review by the human team. This creates an essential quality gate between idea generation and real-world deployment, ensuring the human team retains final operational responsibility.

07 — TAGUCHI ROBUST DESIGN & DOE

A successful experiment under one condition does not guarantee consistent performance elsewhere. Taguchi's **Robust Design** approach is adapted here to build a system capable of maintaining acceptable performance despite operating variation.

Design of Experiments (DOE) systematically tests combinations of key factors rather than relying on isolated trials. The objective is to identify a solution that performs consistently across varied user behaviours and operating environments.

08 — THREE-STAGE CONTROLLED VALIDATION MODEL



09 — FAILURE AS A LEARNING INPUT

If a pilot does not produce expected results, deployment halts and failure becomes an input for root-cause analysis:

Pilot Failure → Root-Cause Analysis → AI Reassessment → Revised Plan → Re-test

The revised solution passes through controlled validation again. Failed attempts generate structural learning rather than repeated operational failures.

10 — CONTINUOUS QUALITY & KAIZEN

Quality does not end when a solution reaches users. Regular feedback and surveys provide continuous quality signals, reflecting the Japanese **Kaizen** philosophy of incremental improvement:

Feedback → Analysis → Improvement → Validation → Standardisation → Monitoring

The objective is maintaining a process capable of continuously detecting and improving quality.

11 — QUALITY ENGINEERING × AI SUMMARY

The proposed framework unifies core methodologies: **AI/LLM Analysis** for signal interpretation, **Predictive Quality** for early degradation identification, **FMEA** for risk evaluation, **Zero Defects** prevention-oriented thinking, **Taguchi Robust Design & DOE** for systematic experimentation, **PDCA** structure, progressive validation, and **Kaizen**.

KEY QUALITY INSIGHT

"Quality at scale is not achieved by predicting what users want alone. It is achieved by detecting change early, designing robust solutions, preventing implementation failures, validating progressively and continuously improving through evidence."

Disclaimer: This case study is an independent professional research exercise by Jagriti Singh. It is a conceptual application of Quality Engineering methodologies to a large-scale personalised content environment and is not an official Meta project, internal Meta analysis, audit, specification, endorsement, partnership or affiliation. The framework represents independent professional reasoning and proposed methodology.